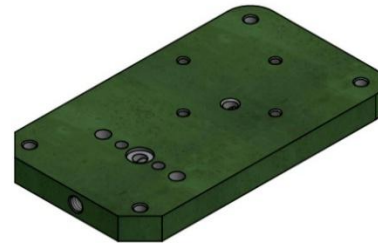
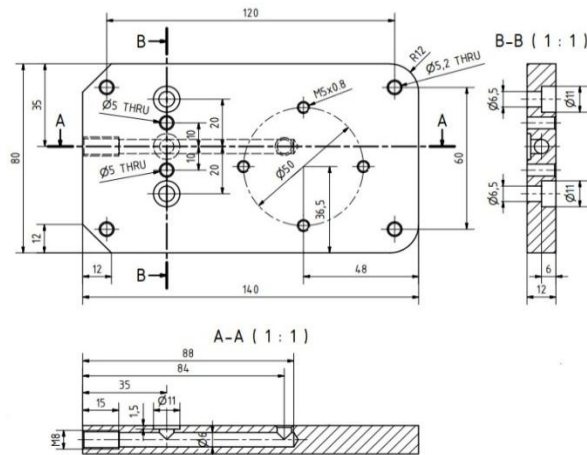


Indian Institute of Technology Mandi
School of Mechanical and Material Engineering
ME205: Machine Drawing
Lab Assignment#17

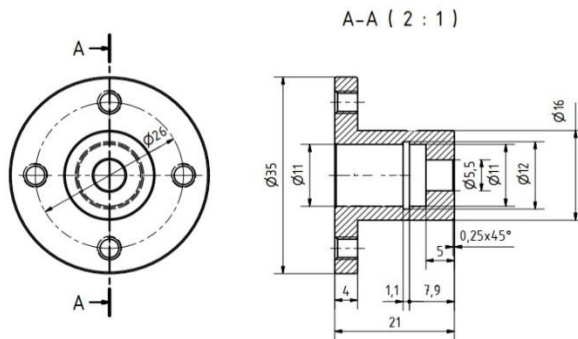
27 March 2026, 02:00 – 04:00 PM

Draw the following figures using CAD software (Solid Works) and Assemble
Please assume suitable dimensions as per requirements
(All Units are in mm)

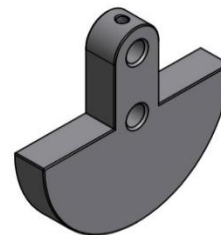
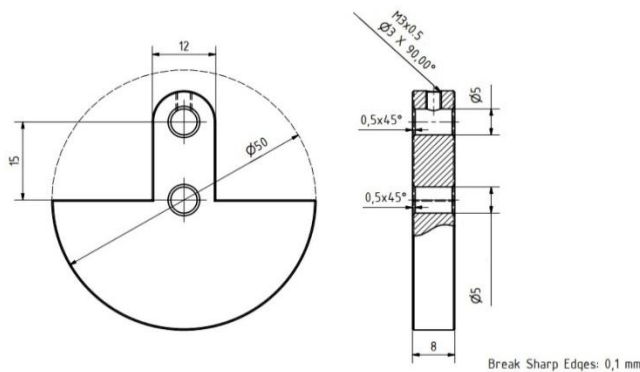
Horizontal Stirling Engine Assembly



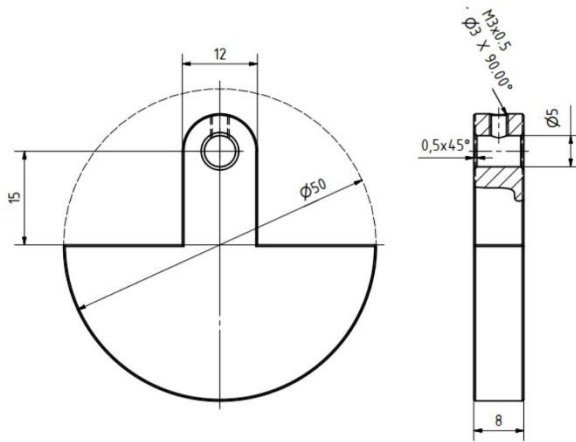
Base



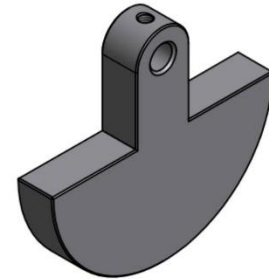
Bearing Support



Counter weight A

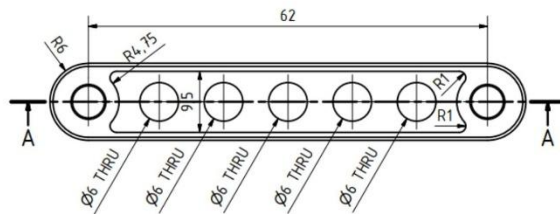
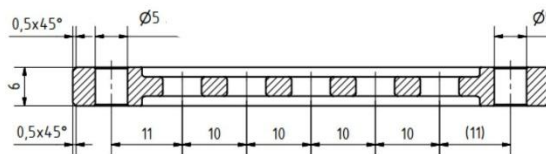


Break Sharp Edges: 0,1 mm

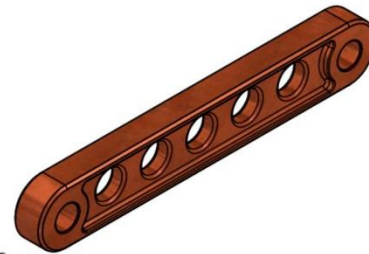


Counter Weight B

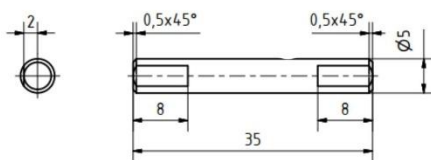
A-A (2 : 1)



Break Sharp Edges: 0,1 mm

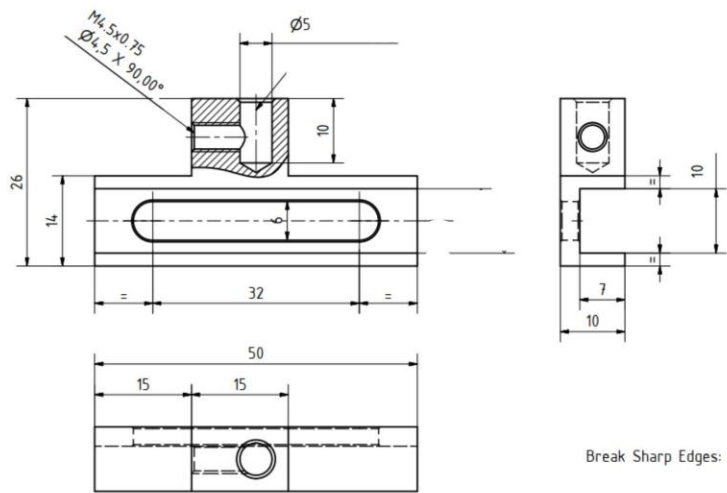


Crank Rod

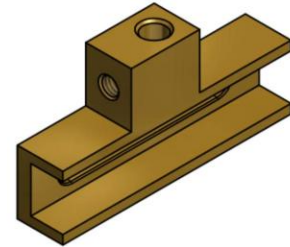


Break Sharp Edges: 0,1 mm

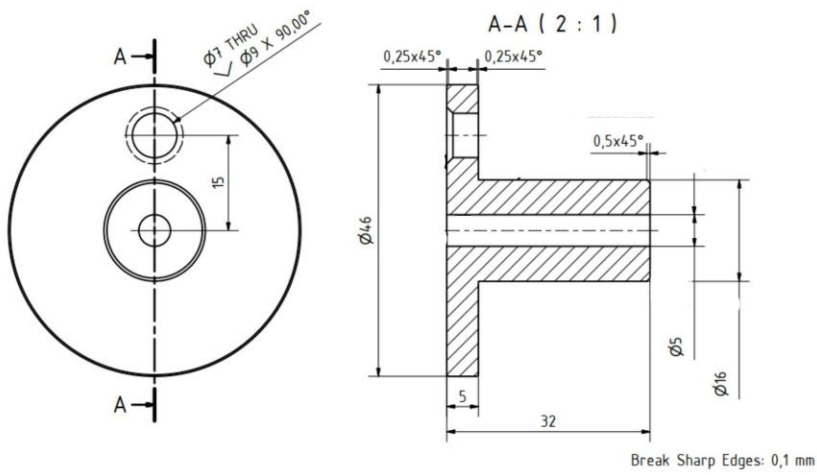
Crank Shaft



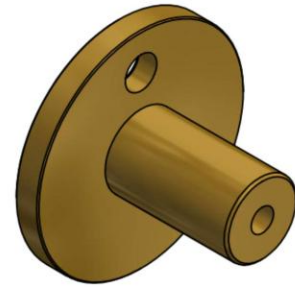
Break Sharp Edges: 0,1 mm



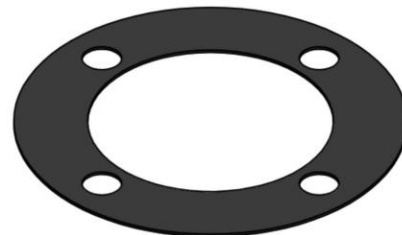
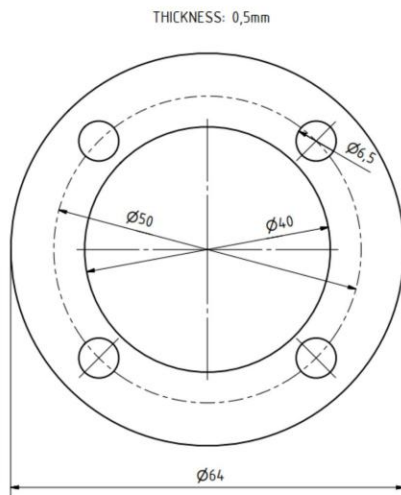
Cross Head



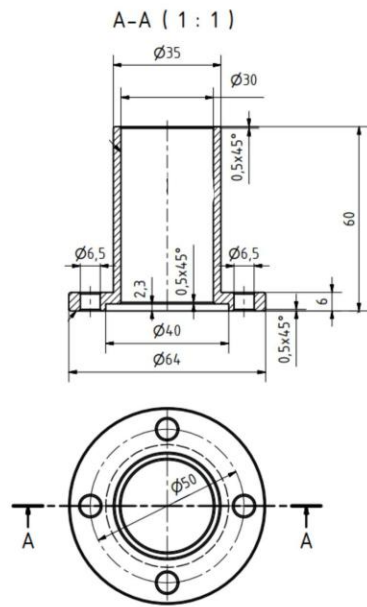
Break Sharp Edges: 0,1 mm



Cylinder Cover

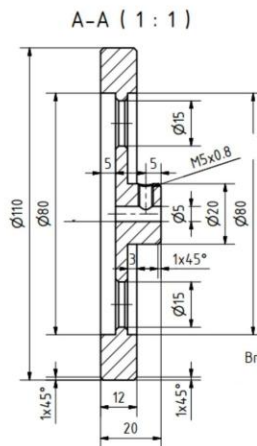
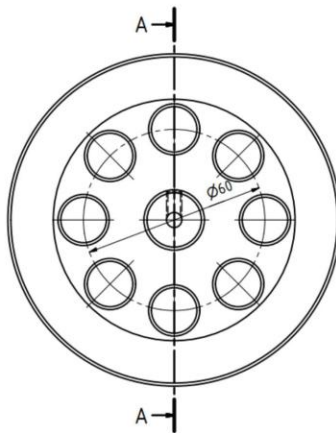


Cylinder Packing



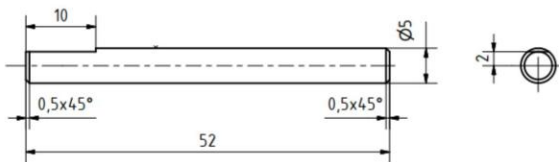
Break Sharp Edges: 0,1 mm

Cylinder



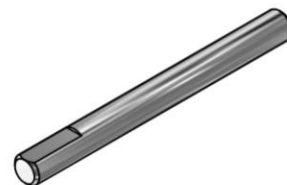
Break Sharp Edges: 0,1 mm

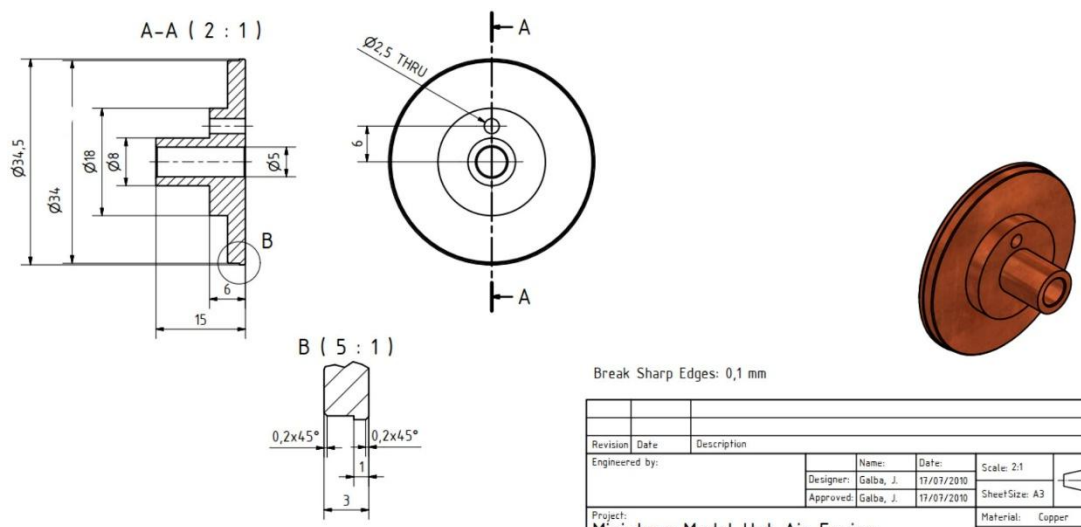
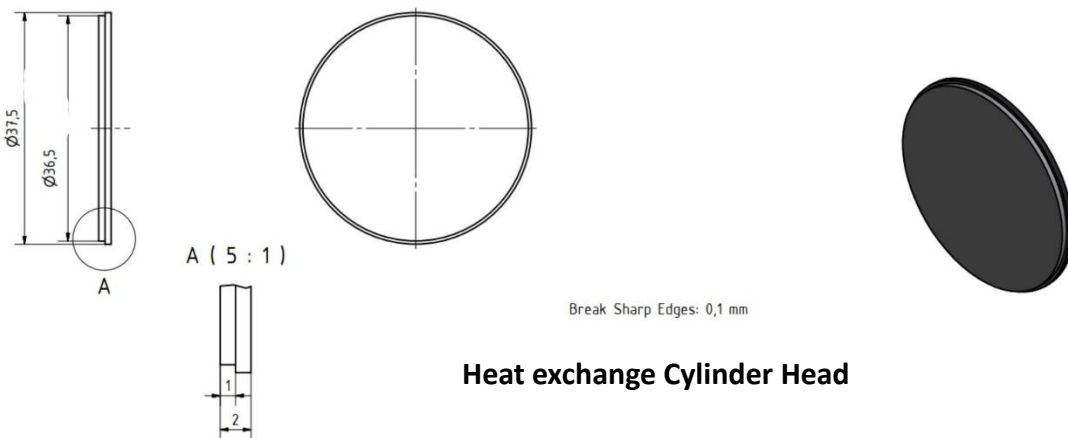
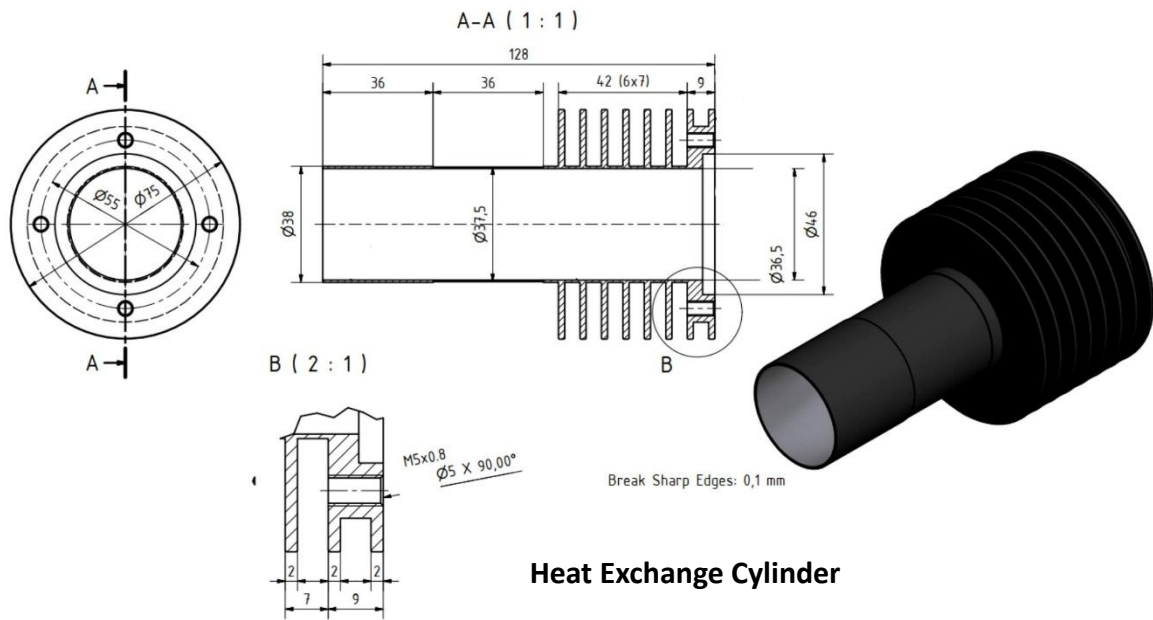
Flywheel



Break Sharp Edges: 0,1 mm

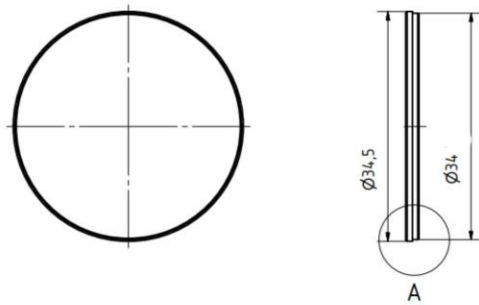
Flywheel shaft



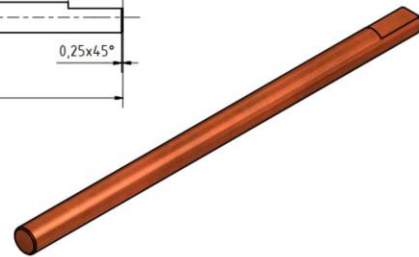
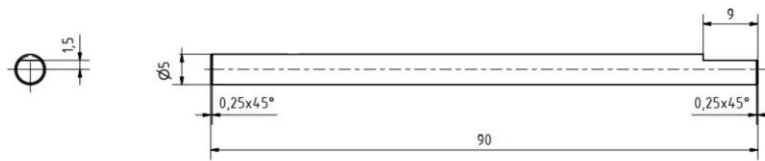
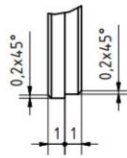


Heat Exchange Piston Exhaust

Heat Exchange Piston Head

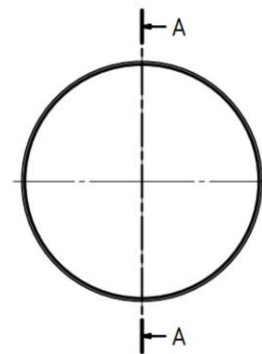
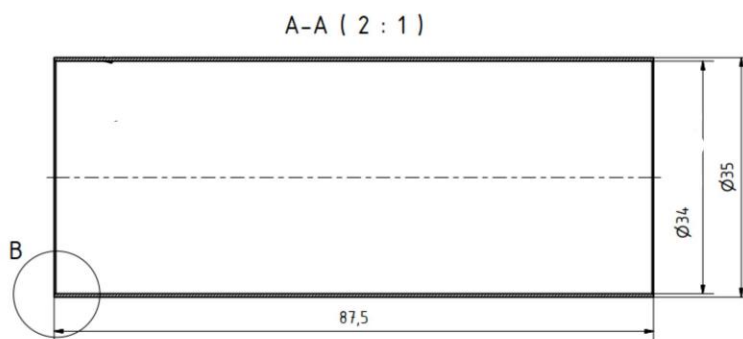


A (5 : 1)

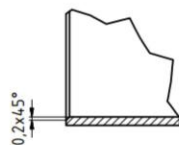


Heat Exchange Piston Shaft

Break Sharp Edges: 0,1 mm

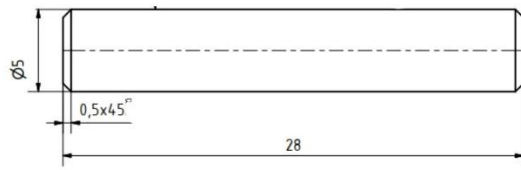


B (5 : 1)

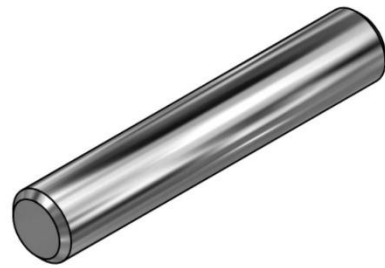


Break Sharp Edges: 0,1 mm

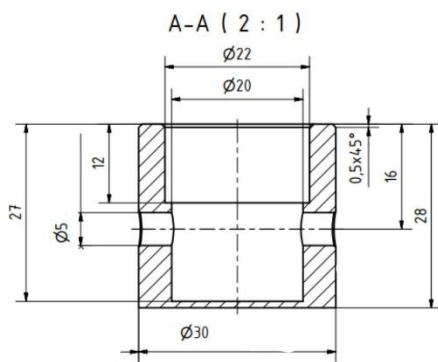
Heat Exchange Piston Tube



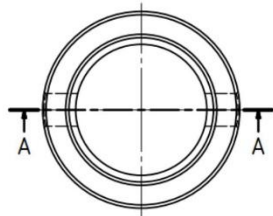
Piston pin



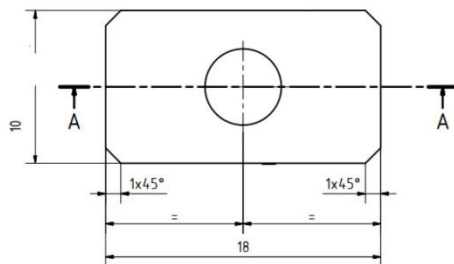
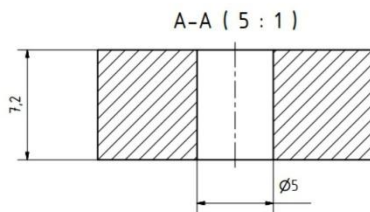
Break Sharp Edges: 0,1 mm



Piston

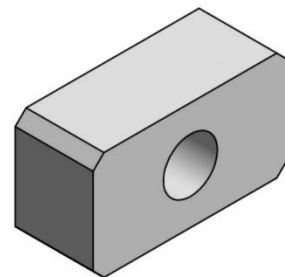


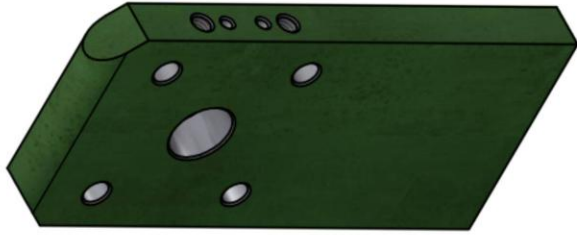
Break Sharp Edges: 0,1 mm



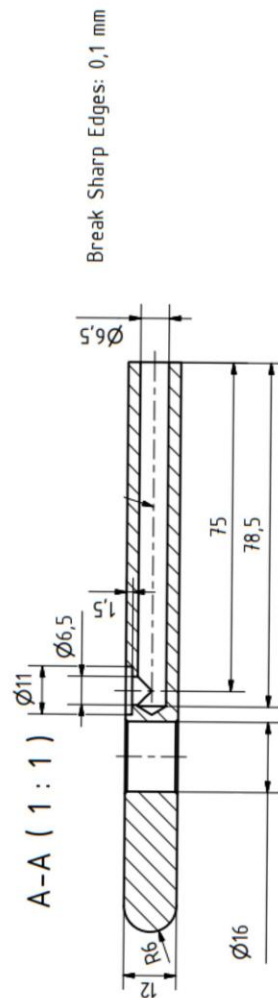
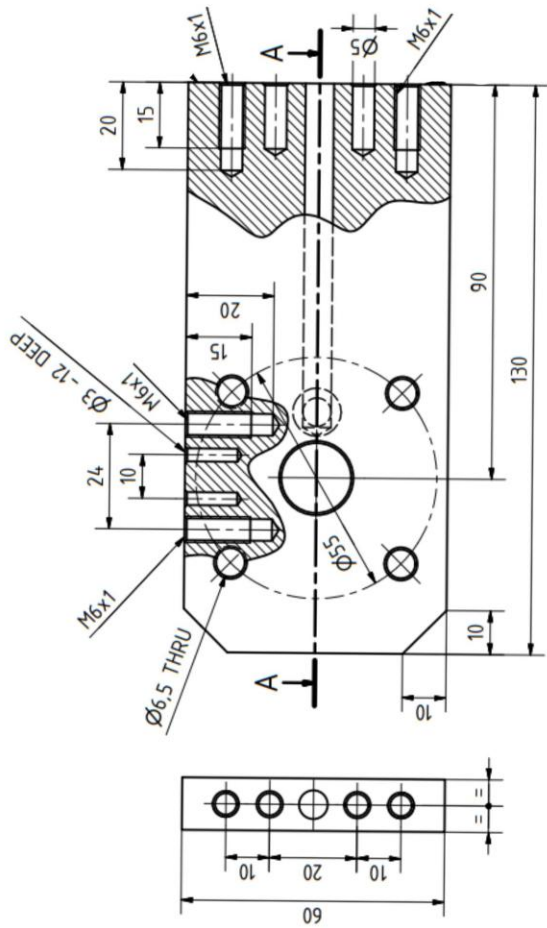
Break Sharp Edges: 0,1 mm

Slide

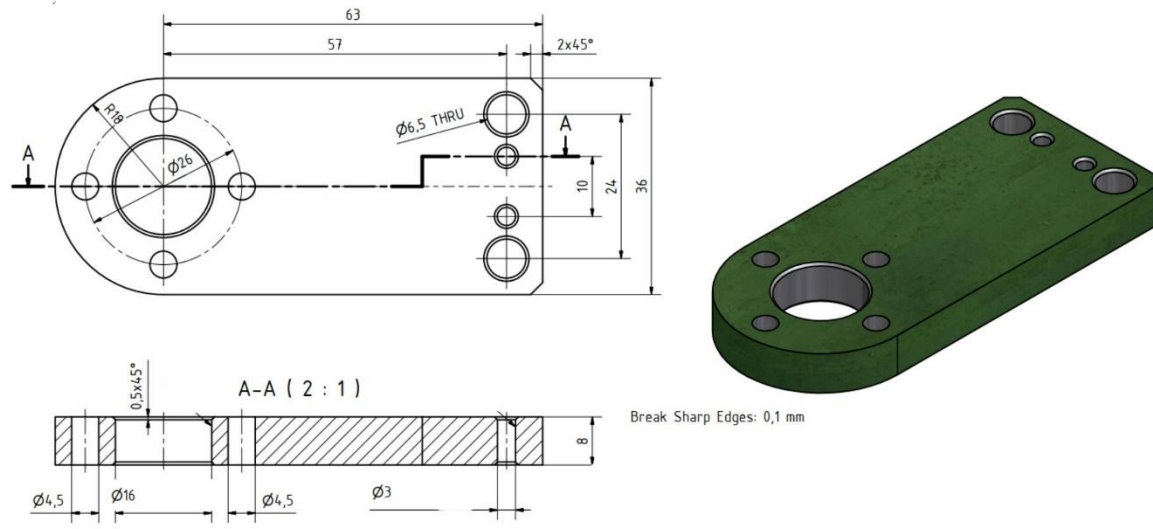




Standard



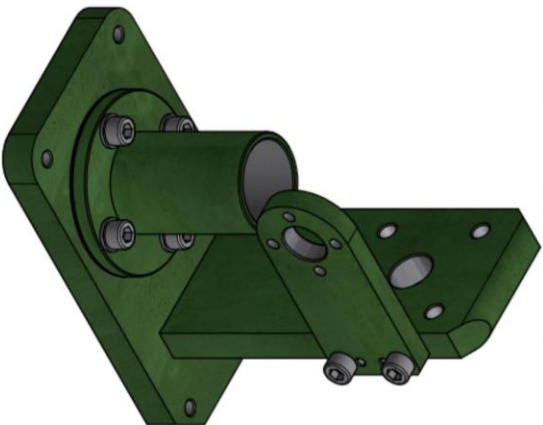
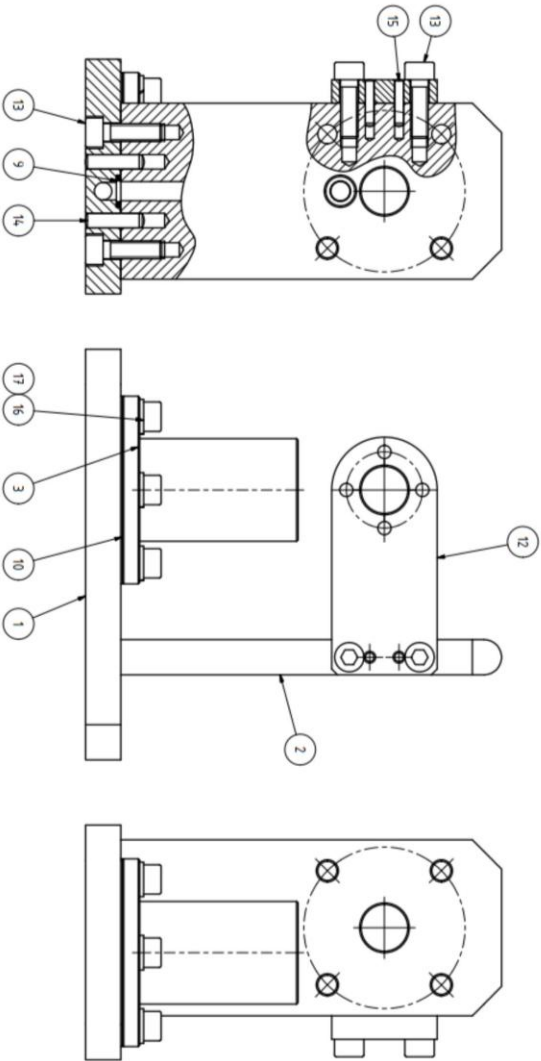
Break Sharp Edges: 0,1 mm



Support

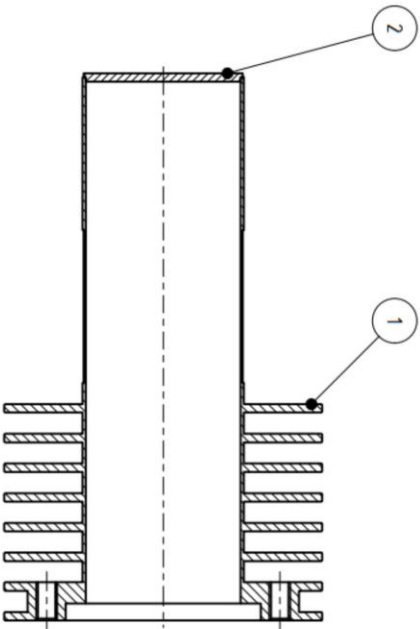
PARTS LIST		
ITEM QTY	PART NUMBER	DESCRIPTION
1	1	Base
2	1	Standard
3	1	Cylinder
9	2	Air Channel Packing
10	1	Cylinder Packing
12	1	Support
13	4	DIN 912 - M6 x 20
14	2	ISO 8734 - 5 x 20
	- A	
15	2	ISO 8734 - 3 x 16
	- A	
16	4	DIN 128 - A6
	- A	
17	4	DIN 912 - M6 x 16

		Cylinder Head Cap Screw
		Parallel pins of hardened steel and
		martensitic stainless steel (Dowel pins)
		Spring Washer
		Cylinder Head Cap Screw



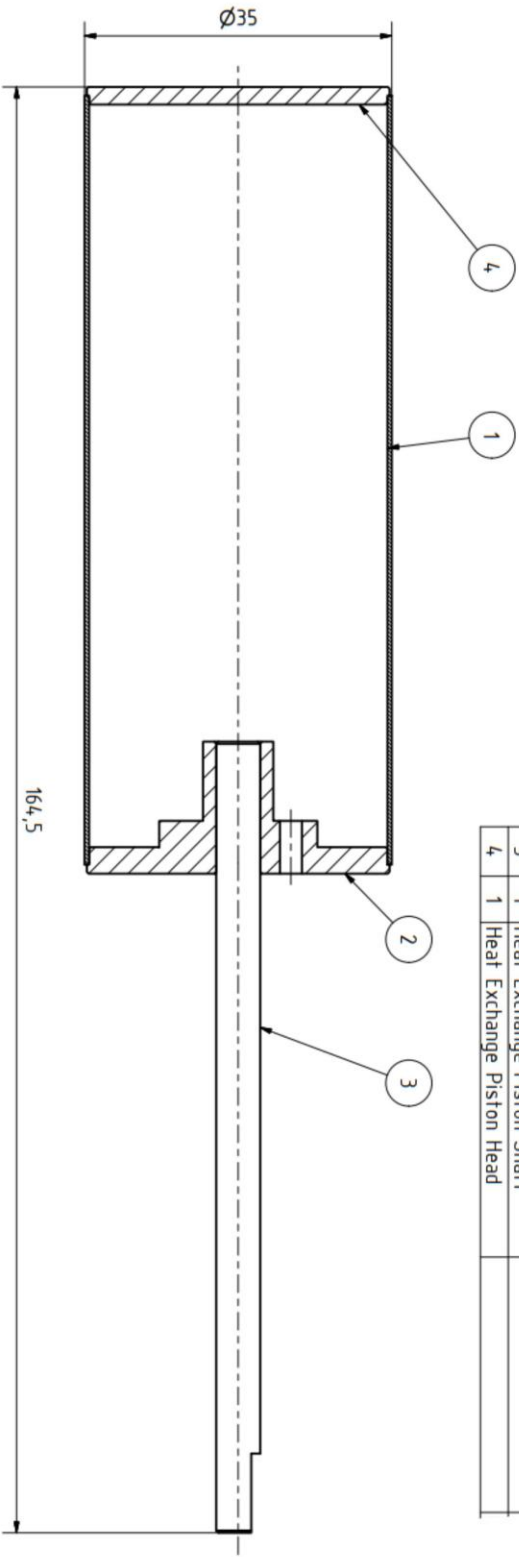
Horizontal Stirling Engine
Assembly Base

ITEM	QTY	PART NUMBER
1	1	Heat Exchange Cylinder
2	1	Heat Exchange Cylinderhead



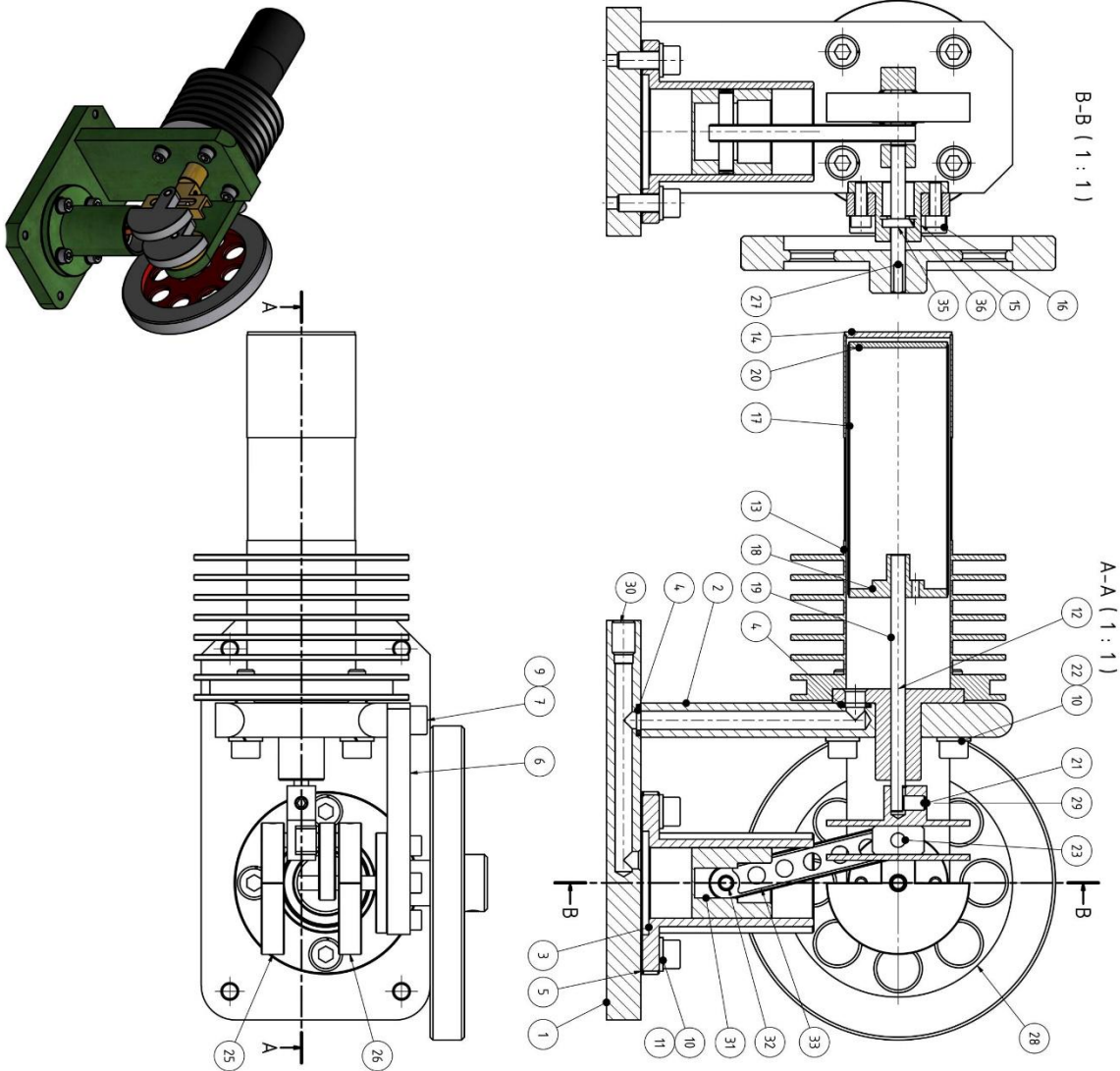
Horizontal Stirling Engine
Assembly Heat Exchange Cylinder

PARTS LIST		
ITEM QTY	PART NUMBER	DESCRIPTION
1 1	Heat Exchange Piston Tube	
2 1	Heat Exchange Piston Exhaust	
3 1	Heat Exchange Piston Shaft	
4 1	Heat Exchange Piston Head	



Miniature Model Hot Air Engine
 Title
 Horizontal Stirling Engine
 Assembly Heat Exchange Piston

Horizontal Stirling Engine Assembly



PARTS LIST		
ITEM QTY	PART NUMBER	DESCRIPTION
1 1	Base	
2 1	Standard	
3 1	Cylinder	
4 2	Air Channel Packing	
5 1	Cylinder Packing	
6 1	Support	
7 4	DIN 912 - M6 x 20	Cylinder Head Cap Screw
8 2	ISO 8734 - 5 x 20 - A	Parallel pins
9 2	ISO 8734 - 3 x 16 - A	Parallel pins
10 8	DIN 128 - A6	Spring Washer
11 4	DIN 912 - M6 x 16	Cylinder Head Cap Screw
12 1	Cylinder Cover	
13 1	Heat Exchange Cylinder	
14 1	Heat Exchange	
15 1	Bearing Support	
16 4	DIN 912 - M4 x 12	Cylinder Head Cap Screw
17 1	Heat Exchange Piston	
18 1	Heat Exchange Piston	
19 1	Heat Exchange Piston	
20 1	Heat Exchange Piston	
21 1	Head	
22 4	DIN 912 - M6 x 25	Cylinder Head Cap Screw
23 1	Slide	
24 1	Grankshaft	
25 1	Counterweight B	
26 1	Counterweight A	
27 1	Flywheel	
28 1	Flywheel	
29 2	DIN 913 - M5 x 8	Hexagon Socket Screw
30 1	DIN 913 - M8 x 12	Hexagon Socket Screw
31 1	Piston	
32 1	Piston Pin	
33 1	Crankrod	
35 1	DIN 625 SKF - SKF 618/5	Single row ball bearings
36 1	DIN 472 - 12x1	Retaining rings for bore
37 2	DIN 913 - M3 x 4	Hexagon Socket Set Screw

Title
Horizontal Stirling Engine
Assembly